SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned working unit — 19 July 2026

Authority. Corrected French lines 2561–2571; original printed pp. 85–86; physical source-PDF p. 76; recomposed running p. 68. The original printed-page marker changes to p. 86 within line 2563. This bounded unit has been independently source-reviewed against the corrected French TeX and direct scan; cumulative integration remains pending. Line 2572 is blank; Proposition 1.4 begins at line 2573 and is excluded.

We shall consider only complexes whose differential has degree $+1$. By the hypothesis on M, there exists a projective resolution of M of finite length,

$$u\colon L^\bullet \longrightarrow M,$$

where, moreover, the L^p are finitely generated modules and $L^p = 0$ if $p \notin [-n, 0]$. On the other hand, let $v\colon M \longrightarrow I^\bullet$ be an injective resolution of M. I claim that

$$v \circ u\colon L^\bullet \longrightarrow I^\bullet \tag{1.1}$$

is an injective resolution of $L^\bullet$. We must specify what this means.

Definition 1.3. — Let $X^\bullet$ be a complex of A-modules. An *injective resolution* of $X^\bullet$ is a homomorphism of complexes

$$x\colon X^\bullet \longrightarrow CX^\bullet$$

such that CX^p is injective for every $p \in \mathbb{Z}$, and x induces an isomorphism in homology.